

# NAdam#

<https://pytorch.org/docs/stable/generated/torch.optim.NAdam.html#torch.optim>

## Description:

Implements NAdam algorithm. For further details regarding the algorithm we refer to Incorporating Nesterov Momentum into Adam. `params` (iterable) – iterable of parameters or `named_parameters` to optimize or iterable of dicts defining parameter groups. When using `named_parameters`, all parameters in all groups should be named `lr` (float, Tensor, optional) – learning rate (default:  $2e-3$ ) `betas` (Tuple[float, float], optional) – coefficients used for computing running averages of gradient and its square (default: (0.9, 0.999))

## Function Signature

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```
class torch.optim.NAdam(params, lr=0.002, betas=(0.9, 0.999), eps=1e-08, weight_decay=0, momentum_decay=0.004, decoupled_weight_decay=False, *, foreach=None, maximize=False, capturable=False, differentiable=False) [source]#
```

Parameters

```
add_param_group(param_group) [source]#
```

Parameters

```
load_state_dict(state_dict) [source]#
```

### Returns:

dict[str, Any]

## Code Examples

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```
>>> model = torch.nn.Linear(10, 10)
>>> optim = torch.optim.SGD(model.parameters(), lr=3e-4)
>>> scheduler1 = torch.optim.lr_scheduler.LinearLR(
...     optim,
...     start_factor=0.1,
...     end_factor=1,
...     total_iters=20,
... )
>>> scheduler2 = torch.optim.lr_scheduler.CosineAnnealingLR(
...     optim,
...     T_max=80,
...     eta_min=3e-5,
... )
>>> lr = torch.optim.lr_scheduler.SequentialLR(
...     optim,
...     schedulers=[scheduler1, scheduler2],
...     milestones=[20],
... )
>>> lr.load_state_dict(torch.load("./save_seq.pt"))
>>> # now load the optimizer checkpoint after loading the
LRScheduler
>>> optim.load_state_dict(torch.load("./save_optim.pt"))
```

```
hook(optimizer) -> None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer) -> None
```

```
hook(optimizer, args, kwargs) -> None
```

```
hook(optimizer, args, kwargs) -> None or modified args and  
kwargs
```

```

{
  'state': {
    0: {'momentum_buffer': tensor(...), ...},
    1: {'momentum_buffer': tensor(...), ...},
    2: {'momentum_buffer': tensor(...), ...},
    3: {'momentum_buffer': tensor(...), ...}
  },
  'param_groups': [
    {
      'lr': 0.01,
      'weight_decay': 0,
      ...
      'params': [0]
      'param_names' ['param0'] (optional)
    },
    {
      'lr': 0.001,
      'weight_decay': 0.5,
      ...
      'params': [1, 2, 3]
      'param_names': ['param1', 'layer.weight',
'layer.bias'] (optional)
    }
  ]
}

```

## Notes & Warnings

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### WARNING

Make sure this method is called after initializing `torch.optim.lr_scheduler.LRScheduler`, as calling it beforehand will overwrite the loaded learning rates.

### NOTE

The names of the parameters (if they exist under the “param\_names” key of each param group in `state_dict()`) will not affect the loading process. To use the parameters’ names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom `register_load_state_dict_pre_hook` should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer `param_names` will remain unchanged.

## Detailed Documentation

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### Documentation

Implements NAdam algorithm.

For further details regarding the algorithm we refer to Incorporating Nesterov Momentum into Adam.

params (iterable) – iterable of parameters or named\_parameters to optimize or iterable of dicts defining parameter groups. When using named\_parameters, all parameters in all groups should be named

lr (float, Tensor, optional) – learning rate (default:  $2e-3$ )

betas (Tuple[float, float], optional) – coefficients used for computing running averages of gradient and its square (default: (0.9, 0.999))

eps (float, optional) – term added to the denominator to improve numerical stability (default:  $1e-8$ )

weight\_decay (float, optional) – weight decay (L2 penalty) (default: 0)

momentum\_decay (float, optional) – momentum momentum\_decay (default:  $4e-3$ )

decoupled\_weight\_decay (bool, optional) – whether to decouple the weight decay as in AdamW to obtain NAdamW. If True, the algorithm does not accumulate weight decay in the momentum nor variance. (default: False)

foreach (bool, optional) – whether foreach implementation of optimizer is used. If unspecified by the user (so foreach is None), we will try to use foreach over the for-loop implementation on CUDA, since it is usually significantly more performant. Note that the foreach implementation uses  $\sim \text{sizeof}(\text{params})$  more peak memory than the for-loop version due to the intermediates being a tensorlist vs just one tensor. If memory is prohibitive, batch fewer parameters through the optimizer at a time or switch this flag to False (default: None)

maximize (bool, optional) – maximize the objective with respect to the params, instead of minimizing (default: False)

capturable (bool, optional) – whether this instance is safe to capture in a graph, whether

for CUDA graphs or for `torch.compile` support. Tensors are only capturable when on supported accelerators. Passing `True` can impair ungraphed performance, so if you don't intend to graph capture this instance, leave it `False` (default: `False`)

`differentiable` (bool, optional) – whether autograd should occur through the optimizer step in training. Otherwise, the `step()` function runs in a `torch.no_grad()` context. Setting to `True` can impair performance, so leave it `False` if you don't intend to run autograd through this instance (default: `False`)

Add a param group to the Optimizer's `param_groups`.

This can be useful when fine tuning a pre-trained network as frozen layers can be made trainable and added to the Optimizer as training progresses.

`param_group` (dict) – Specifies what Tensors should be optimized along with group specific optimization options.

Load the optimizer state.

`state_dict` (dict) – optimizer state. Should be an object returned from a call to `state_dict()`.

Make sure this method is called after initializing `torch.optim.lr_scheduler.LRScheduler`, as calling it beforehand will overwrite the loaded learning rates.

The names of the parameters (if they exist under the “`param_names`” key of each param group in `state_dict()`) will not affect the loading process. To use the parameters' names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom `register_load_state_dict_pre_hook` should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer

param\_names will remain unchanged.

Register a load\_state\_dict post-hook which will be called after load\_state\_dict() is called. It should have the following signature:

The optimizer argument is the optimizer instance being used.

The hook will be called with argument self after calling load\_state\_dict on self. The registered hook can be used to perform post-processing after load\_state\_dict has loaded the state\_dict.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided post hook will be fired before all the already registered post-hooks on load\_state\_dict. Otherwise, the provided hook will be fired after all the already registered post-hooks. (default: False)

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemoveableHandle

Register a load\_state\_dict pre-hook which will be called before load\_state\_dict() is called. It should have the following signature:

The optimizer argument is the optimizer instance being used and the state\_dict argument is a shallow copy of the state\_dict the user passed in to load\_state\_dict. The hook may modify the state\_dict inplace or optionally return a new one. If a state\_dict is returned, it will be used to be loaded into the optimizer.

The hook will be called with argument self and state\_dict before calling load\_state\_dict on self. The registered hook can be used to perform pre-processing before the load\_state\_dict call is made.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided pre hook will be fired before all the already registered pre-hooks on load\_state\_dict. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: False)

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemoveableHandle

Register a state dict post-hook which will be called after state\_dict() is called.

It should have the following signature:

The hook will be called with arguments self and state\_dict after generating a state\_dict on self. The hook may modify the state\_dict inplace or optionally return a new one. The registered hook can be used to perform post-processing on the state\_dict before it is returned.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided post hook will be fired before all the already registered post-hooks on state\_dict. Otherwise, the provided hook will be fired after all the already registered post-hooks. (default: False)

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemoveableHandle

Register a state dict pre-hook which will be called before state\_dict() is called.

It should have the following signature:

The optimizer argument is the optimizer instance being used. The hook will be called with argument self before calling state\_dict on self. The registered hook can be used to perform pre-processing before the state\_dict call is made.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided pre hook will be fired before all the already registered pre-hooks on state\_dict. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: False)

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemoveableHandle

Register an optimizer step post hook which will be called after optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling handle.remove()

torch.utils.hooks.RemovableHandle

Register an optimizer step pre hook which will be called before optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used. If args and kwargs are modified by the pre-hook, then the transformed values are returned as a tuple containing the new\_args and new\_kwargs.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemovableHandle`

Return the state of the optimizer as a dict.

It contains two entries:

differs between optimizer classes, but some common characteristics hold. For example, state is saved per parameter, and the parameter itself is NOT saved. state is a Dictionary mapping parameter ids to a Dict with state corresponding to each parameter.

parameter group is a Dict. Each parameter group contains metadata specific to the optimizer, such as learning rate and weight decay, as well as a List of parameter IDs of the parameters in the group. If a param group was initialized with `named_parameters()` the names content will also be saved in the state dict.

NOTE: The parameter IDs may look like indices but they are just IDs associating state with `param_group`. When loading from a `state_dict`, the optimizer will zip the `param_group` params (int IDs) and the optimizer `param_groups` (actual `nn.Parameter s`) in order to match state WITHOUT additional verification.

A returned state dict might look something like:

Perform a single optimization step.

closure (Callable, optional) – A closure that reevaluates the model and returns the loss.

Reset the gradients of all optimized `torch.Tensor s`.

`set_to_none` (bool, optional) –

Instead of setting to zero, set the grads to None. Default: True

This will in general have lower memory footprint, and can modestly improve performance. However, it changes certain behaviors. For example:

When the user tries to access a gradient and perform manual ops on it, a None attribute or a Tensor full of 0s will behave differently.

If the user requests `zero_grad(set_to_none=True)` followed by a backward pass, `.grads` are guaranteed to be None for params that did not receive a gradient.

`torch.optim` optimizers have a different behavior if the gradient is 0 or None (in one case it does the step with a gradient of 0 and in the other it skips the step altogether).